

General information:	3-0-0 = 3 GTU credits, 5 ECTS credits, Th.: 13:00-16:00, (Face-to-Face Harita Z06)	
Instructor:	Assist. Prof. Saeed LOTFAN H2 123, slotfan@gtu.edu.tr, (262) 605 26 58	
Teaching assistants:	Mustafa KEMAL ACAR H2 Z42, mkemalacar@gtu.edu.tr, (262) 605 10 00	
Office hours:	Th.: 10:00-13:00	
Website:	Information related to the course (announcements, files, assignments, etc.) will be shared via MS Teams. It is the student's responsibility to regularly check the relevant system(s). In case of any technical issues with these systems, communication will be carried out through GTU email accounts. General ME 331 SD&V Fall 2026 Microsoft Teams	
Textbooks:	1. S. S. Rao, Mechanical Vibrations, Pearson, 2017. 2. L. Meirowitch, Fundamentals of Vibrations, McGraw Hill, 2001. 3. Thomson, William. Theory of vibration with applications. CrC Press, 2018. 4. D. J. Inman, Engineering Vibration, 4th Edition, Pearson, 2014. 5. S. G. Kelly, Fundamentals of Mechanical Vibrations, McGraw Hill, 1993.	
Catalog description:	It is aimed at teaching the fundamental concepts of how systems vibrate. Basic aspects of vibrations for mathematical modeling, derivation/solution of equations of motion, and subsequent system analysis will be covered for discrete systems.	
Prerequisite:	ME 232 (Minimum DD)	
Course type:	This course is required for all Mechanical Engineering students.	
Integrity statement:	All students are expected to act with civility, academic and personal integrity, respect others' dignity, rights, and property; and help create and maintain an environment in which all can succeed through the fruits of their own efforts.	
Grading:	Midterm	25%
	Quiz	10%
	HomeWorks	10%
	Project	30%
	Final exam	25%
	a. Attendance is mandatory: at least 70% for theoretical classes and at least 80% for laboratory or practical sessions. Students who do not meet the attendance requirement will receive an NA grade regardless of their performance during the semester and will not be allowed to take the final exam. b. A student who fails to achieve at least 30% success in the weighted average of midterm exam and quizzes evaluations during the term will fail with a VF grade; they are not entitled to take the final exam.	

- c. Students whose success in the final exam is below **25/100** are considered to have failed the course with an **FF** grade, regardless of their other studies.

Not failing the items a, b and c above does not mean to have succeeded in the course; for students who meet the minimum requirements stated in these articles, course grade evaluation is made separately in the [FF-AA] range.

Other important notes:

- Students who receive a VF or NA grade are not allowed to take the final exam.
- For tracking course attendance, the “FR-0345 Course Attendance Sheet” and for reporting absences, the “FR-0349 Student Absence Notification” will be used.
- All exams must be completed without receiving or providing assistance to others. If cheating is detected, disciplinary action will be taken in accordance with university regulations.
- For matters not specified in this document regarding the requirements for attending quizzes, midterm exams, and final exams, as well as exam evaluation and eligibility for make-up exams, the general rules stated in the undergraduate regulations shall apply.
- Exams and related procedures will be conducted in accordance with the most recent guidelines issued by the University Senate, the Faculty of Engineering, or the Mechanical Engineering Department. Any rules in this document that become outdated due to such updates shall be considered invalid.

The following table will be taken into account in the calculation of the final grade.

Score	Letter Grade
88-100	AA
79-87	BA
70-78	BB
61-69	CB
53-60	CC
46-52	DC
40-45	DD
The conditions for receiving an FF grade are given above.	FF
The conditions for receiving a VF grade are given above.	VF
The conditions for receiving an NA grade are given above.	NA

Expected Workload:

It is three hours before each class and three hours for each assignment. For students who received a grade lower than BB in the prerequisite, this time may exceed an average of six hours per week.

Topics covered and Schedule:

Week	Date	Topic	
1	24-Sep.	Introduction to vibration; degrees of freedom; mass, spring, and damping models	
2	01-Oct.	Equivalent elements, harmonic motion, energy, and small-motion modeling	
3	08-Oct.	Undamped free SDOF vibration; natural frequency and Rayleigh method	
4	15-Oct.	Damped free SDOF vibration; viscous, Coulomb, and hysteretic damping	
5	22-Oct.	Harmonic force response; resonance, amplitude, and phase	
6	29-Oct.	Base excitation, transmissibility, and rotating unbalance (makeup; format/date TBA)	
7	05-Nov.	General forces I: Fourier series, impulse, and step response	
8	12-Nov.	Midterm exam; review of modeling and SDOF response from prior weeks	M
9	19-Nov.	Midterm feedback; general forces II: convolution and numerical response	
10	26-Nov.	Free and forced 2DOF vibration; modes and coordinate coupling	
11	03-Dec.	MDOF modeling and matrix equations of motion	
12	10-Dec.	MDOF natural frequencies, mode shapes, and orthogonality	
13	17-Dec.	Forced MDOF response using modal superposition	
14	24-Dec.	Approximate natural frequencies; vibration measurement and applications	
Makeup	TBA	Delivery of week 6 content, possibly online; details to be announced	
Final	04-15 Jan.	Final exam during university exam period; exact date and room TBA	F